

# Building Robust Multimodal Sentiment Recognition via a Simple yet Effective Multimodal Transformer

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## ABSTRACT

In this paper, we present the solutions to the MER-MULTI and MER-NOISE sub-challenges of the Multimodal Emotion Recognition Challenge (MER 2023). For the tasks MER-MULTI and MER-NOISE, participants are required to recognize both discrete and dimensional emotions. Particularly, in MER-NOISE, the test videos are corrupted with noise, necessitating the consideration of modality robustness. Our empirical findings indicate that different modalities contribute differently to the tasks, with a significant impact from the audio and visual modalities, while the text modality plays a weaker role in emotion prediction. To facilitate subsequent multimodal fusion, and considering that language information is implicitly embedded in large pre-trained speech models, we have made the deliberate choice to abandon the text modality and solely utilize visual and acoustic modalities for these sub-challenges. To address the potential underfitting of individual modalities during multimodal training, we propose to jointly train all modalities via a weighted blending of supervision signals. Furthermore, to enhance the robustness of our model, we employ a range of data augmentation techniques at the *image level*, *waveform level*, and *spectrogram level*. Experimental results show that our model **ranks 1st** in both MER-MULTI (0.7005) and MER-NOISE (0.6846) sub-challenges, validating the effectiveness of our method. Our code is publicly available at <https://github.com/dingchaoyue/Multimodal-Emotion-Recognition-MER-and-MuSe-2023-Challenges>.

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MM '23, October 29–November 3, 2023, Ottawa, ON, Canada.

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ACM ISBN 979-8-4007-0108-5/23/10...\$15.00  
<https://doi.org/10.1145/3581783.3612872>

## CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; • **Information systems** → *Multimedia information systems*; **Sentiment analysis**.

## KEYWORDS

Multimodal Sentiment Analysis; Multimodal Fusion; Modality Robustness

### ACM Reference Format:

\*Daoming Zong, \*Chaoyue Ding, †Baoxiang Li, Dinghao Zhou, Jiakui Li, Ken Zheng, and Qunyan Zhou. 2023. Building Robust Multimodal Sentiment Recognition via a Simple yet Effective Multimodal Transformer. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3581783.3612872>

## 1 INTRODUCTION

Sentiment analysis holds significant importance in human-computer interaction as it facilitates systems in comprehending and responding to users' emotional states [12]. Emotion expression encompasses diverse modalities, including facial expressions, speech, posture, physiological signals, and language [17]. The fundamental task of emotion recognition involves the integration of information from multiple modalities to accurately infer and comprehend the underlying sentiment. This integration poses a significant challenge, requiring sophisticated approaches that can effectively fuse and leverage multimodal cues for improved emotion recognition performance. Recently, transformer-based video emotion recognition [4, 19, 30] has attracted widespread attention due to the evolution of vision transformer [9] and audio transformer [1, 13].

Research in multimodal machine learning can be broadly classified into three fusion strategies: *early fusion*, *intermediate fusion* (a.k.a. *mid fusion*), and *late fusion* [2]. Early works in multimodal sentiment analysis predominantly focused on early or late fusion techniques with attention [7, 12, 28, 29]. More recently, there has been a surge of interest in exploring attention-based [6, 8, 31] intermediate

fusion approaches [22, 23], particularly utilizing cross-attention [3]. However, it has been observed that multimodal fusion sometimes yields less effective results compared to unimodal approaches. This can be attributed to issues such as overfitting [12, 14, 24], where the model becomes overly reliant on one modality [15], or insufficient training of individual modalities. Additionally, researchers have also explored the efficiency of multimodal fusion [21–23, 27]. Several notable works, such as MBT [23] and EMT [27]. Particularly, PMR [22] introduced the concept of a common message hub to facilitate the effective capturing of cross-modal dynamics in an efficient manner. These approaches allow all modalities to interact solely with the shared proxy modality, enabling more efficient and effective multimodal fusion.

MER-MULTI and MER-NOISE refer to the task of recognizing both discrete emotions and dimensional emotions (valences). The discrete emotion label set consists of *neutral*, *anger*, *happiness*, *sadness*, *worry*, and *surprise*. Valence represents the degree of pleasure, with values ranging from small to large indicating a sentiment shift from negative to positive. Negative emotions such as *anger*, *sadness*, and *worry* typically exhibit valence scores below 0, whereas positive emotions like *happiness* tend to have valence scores predominantly above 0. Neutral emotional states align with a valence score around 0. The emotion of surprise, being multifaceted, encompasses various nuanced meanings such as sadly surprised, angrily surprised, or happily surprised. As a result, the valence for surprise emotions spans the range from negative to positive. In MER-NOISE, video-level corruption is introduced in the test set. For the audio modality, noises from speech, extracted from a subset of MUSAN [25], are added to the clean audio based on a specified speech-to-noise ratio (SNR) ranging from 5 to 10. In the visual modality, blurring is applied to each video frame by downsampling the image to reduce details and subsequently upsampling the resulting low-resolution frame to maintain the original size.

We begin by analyzing the baseline results on MER-MULTI and MER-NOISE. Unimodal results of the baseline reveal: i) both MER-MULTI and MER-NOISE exhibit preferences towards the audio and visual modalities. Specifically, the audio modality achieves optimal performance in MER-MULTI, while the visual modality demonstrates the best performance in MER-NOISE. However, the text modality consistently underperforms in both tasks. The baseline employs a late fusion approach where all modality representations are concatenated, and attention scores are computed for each modality. The multimodal features are obtained by calculating a weighted sum of the concatenated representations using the attention scores. The baseline multimodal results show that both bimodal and trimodal fusions improve performance, outperforming any single unimodal result. However, this simple multimodal fusion approach may lead to modalities competing [15] with each other, resulting in only a subset of modalities being adequately trained and the true potential of the model remaining untapped. This motivates us to develop a more effective multimodal training strategy that ensures sufficient training for each modality, thereby enhancing the model's generalization capability. To sum up, the main contributions of this work are as follows:

- We adopt a simple yet effective Video-Audio Transformer, dubbed VAT, which allows efficient unimodal learning and multimodal interactions across multiple transformer layers.
- We propose a Task-Aware Adaptive Training (TAAT) strategy that enables joint training of multiple modalities by assigning separate optimizers based on the convergence speed of each modality and employing weighted blending of supervision signals. The weights assigned to the supervision signals for each modality are determined based on their respective contributions to the task.
- We employ diverse data augmentation techniques for robust model training, including offline image blurring, on-the-fly environmental corruptions using MUSAN [25] speech noise, and on-the-fly frequency and time domain SpecAugment with speed augmentation. Experimental results demonstrate that these data augmentation strategies significantly enhance the robustness of our model.

## 2 APPROACH

### 2.1 Feature Extraction

In order to achieve good performance in sentiment emoticon recognition (SER), it is crucial to extract fine-grained multimodal features. In this section, we elaborate on the feature extraction methods employed for the three modalities.

**2.1.1 Visual Modality.** Following the official baseline [5], we use all face images extracted from the OpenFace [34]<sup>1</sup> toolkit. The pre-trained MANet-RAFDB [18, 35]<sup>2</sup> model is applied to extract frame-level visual features, leading to a sequence of frame-level visual representations  $\mathbf{H}_v \in \mathbb{R}^{N_v \times 1024}$  for each given video clip.

**2.1.2 Acoustic Modality.** We extract features for a raw audio input by leveraging a large Chinese version of HuBERT [13]<sup>3</sup>, pretrained on WenetSpeech [33] that contains 10,000 hours of Chinese data, and use the averaged hidden representations across the last four layers of the Chinese-HuBERT-Large, resulting in a sequence of audio representations  $\mathbf{H}_a \in \mathbb{R}^{N_a \times 1024}$  for each audio input.

**2.1.3 Textual Modality.** As analyzed in Sec. 1, we made a deliberate choice to exclude the text modality in these two tasks. The rationale behind this decision is based on the premise that the language information is already encoded in Chinese-HuBERT-Large, and incorporating the textual modality may cause additional competition between modalities [15], making the model under-fitting.

### 2.2 Model Architecture

In this section, we first introduce the proposed video-to-audio converter (VAT). We then describe the data augmentation strategy we devised to enhance the robustness of the model.

**2.2.1 Video-Audio Transformer (VAT).** As illustrated in Fig. 1, VAT contains a video encoder, an audio encoder, and a multimodal encoder. We stack 6 layers of standard transformer blocks [31] as a

<sup>1</sup><https://github.com/cmusatyalab/openface>

<sup>2</sup>[https://drive.google.com/file/d/1wT2h5sz22SaEL4YTBwTIB3WoL4HUvg5B/view?usp=share\\_link](https://drive.google.com/file/d/1wT2h5sz22SaEL4YTBwTIB3WoL4HUvg5B/view?usp=share_link)

<sup>3</sup><https://huggingface.co/TencentGameMate/chinese-hubert-large>

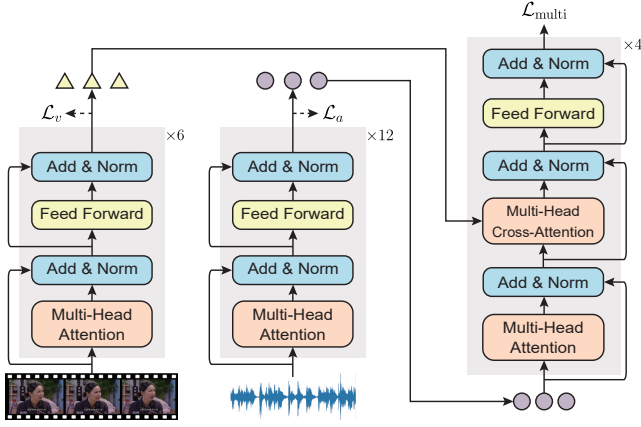


Figure 1: Illustration of our Video-Audio Transformer (VAT).

video encoder. The input to the video encoder is a sequence of facial visual features extracted from the pretrained MANet [35]. The audio encoder is initialized using the Chinese-HuBERT-Large [13] model, and the input to the audio encoder is the raw audio signals. The multimodal encoder is initialized using the last 4 layers of the Chinese-HuBERT-Large model. The audio encoder transforms an input audio into a sequence of embeddings  $\mathbf{H}_a = \{a_{cls}, a_1, \dots, a_{N_a}\}$ , which is fed to the multimodal encoder. The video features  $\mathbf{H}_v = \{v_{cls}, v_1, \dots, v_{N_v}\}$  are fused with the speech features through cross attention at each layer of the multimodal encoder.

**2.2.2 Data Augmentation.** In order to enhance the robustness of our model, we have incorporated various data augmentation techniques at different levels of the input data, including the image level, waveform level, and spectrogram level. At the image level, we apply image blurring. At the waveform level, we employ techniques such as time stretching, pitch shifting, and background noise addition using the speech noise from MUSAN [25]. At the spectrogram level, we apply techniques such as time warping and frequency masking.

### 2.3 Task-Aware Adaptive Training Strategy

Multimodal joint training holds the potential to leverage cross-modal interactions and adequately train individual modalities. However, a simple late fusion approach with a single optimization strategy may lead to insufficient training of the individual modalities, compromising the overall generalization capability of the model. To ensure effective and sufficient training for each modality, we adopt a joint training approach where two modalities are trained together using a weighted blending of supervision signals.

To determine the weight value for each modality, we consider both the model's convergence rate and generalization rate, similar to [32]. Specifically, we define the generalization rate  $G$  as follows:

$$G_n = (\mathcal{L}_{dev}(\Theta^{[0]}) - \mathcal{L}_{dev}(\Theta^{[n]})) \quad (1)$$

where  $\mathcal{L}_{dev}$  represents the held-out validation loss on the validation set.  $\Theta^{[i]}$  indicates the model parameters at the  $i$ -th iteration. Additionally, we define the convergence rate  $C$  as follows:

$$C_n = (\mathcal{L}_{train}(\Theta^{[0]}) - \mathcal{L}_{train}(\Theta^{[n]})) - (\mathcal{L}_{dev}(\Theta^{[0]}) - \mathcal{L}_{dev}(\Theta^{[n]})) \quad (2)$$

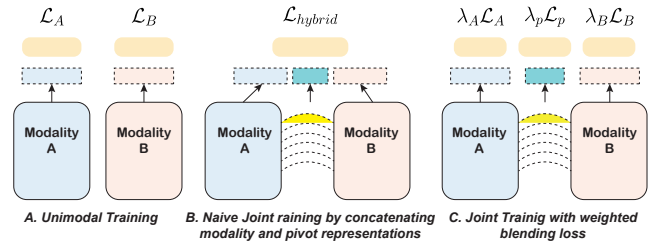


Figure 2: Illustration of different training paradigms. Here we take any two modalities as an example: A) Unimodal training of two different modalities. B) Naïve joint training of two modalities by late fusion. C) Joint training of two modalities with weighted blending of supervision signals.

where  $\mathcal{L}_{train}$  denotes the training loss on the training set.

Our model's training consists of two distinct phases. In Phase I, we conduct separate training for the unimodal encoders and the cross-modal encoder. To achieve this, we attach classifiers to the outputs of the unimodal and cross-modal encoders (as shown in Fig. 2 A and B). Next, we compute the degree of generalization ( $G_{n,j}, j \in \{v, a, fused\}$ ) and the degree of convergence ( $C_{n,j}, j \in \{v, a, fused\}$ ) for the visual, audio, and fused modalities. Based on these calculations, we derive a set of weight factors for each modality, denoted as  $\{\lambda_v, \lambda_a, \lambda_{fused}\} = \left(\frac{1}{\alpha} \left[ \frac{G_{n,v}}{C_{n,v}}, \frac{G_{n,a}}{C_{n,a}}, \frac{G_{n,fused}}{C_{n,fused}} \right]\right)$ . Here  $\alpha$  is a scaling factor. In Phase II, we proceed with joint training of the entire network using a weighted blending of supervision signals, as shown in Fig. 2 C. This entails leveraging the weight factors obtained in Phase I to determine the relative contributions of each modality. By combining the supervised signals from the visual, audio, and fused modalities according to the respective weight factors, we effectively optimize the model during joint training.

During training, we utilize the cross-entropy loss as the classification loss, denoted as  $\mathcal{L}_{cls}$ , and the mean squared error (MSE) as the regression loss, denoted as  $\mathcal{L}_{reg}$ . However, in our experiments, we observed that our model struggles to effectively capture the implicit dependency between discrete emotions and dimensional emotions. To address this issue, we train two separate models for discrete emotions and valence, treating them as independent tasks. Consequently, the optimization goal for Phase II is defined as follows:

$$\mathcal{L}_{emo} = \lambda_v \mathcal{L}_{cls}(e_v, y) + \lambda_a \mathcal{L}_{cls}(e_a, y) + \lambda_{fused} \mathcal{L}_{cls}(e_f, y), \quad (3)$$

$$\mathcal{L}_{val} = \lambda_v \mathcal{L}_{reg}(e_v, y) + \lambda_a \mathcal{L}_{reg}(e_a, y) + \lambda_{fused} \mathcal{L}_{reg}(e_f, y), \quad (4)$$

where  $\mathcal{L}_{emo}$  is the total loss for discrete emotion classification and  $\mathcal{L}_{val}$  is the total loss for valence identification in phase II.

## 3 EXPERIMENTS

### 3.1 Datasets and Evaluation Metric

The dataset statistics for MER-MULTI and MER-NOISE can be found in Table 1. The Train&Val provide annotations for both discrete and dimensional emotions. For the test samples, both MER-MULTI and MER-NOISE aim to predict these emotion labels. In MER-NOISE, video-level noise is randomly added to the test samples, including audio noise and image blurring. For the discrete emotion, the

**Table 1: Statistics of the MER 2023 dataset.**

| Partition | # of samples |           | Duration (hh:mm:ss) |
|-----------|--------------|-----------|---------------------|
|           | labeled      | unlabeled |                     |
| Train&Val | 3373         | 0         | 03:45:47            |
| MER-MULTI | 411          | 0         | 00:28:09            |
| MER-NOISE | 412          | 0         | 00:26:23            |

**Table 2: Performance comparison between baseline models and VAT in unimodal and multimodal settings. Here, ‘A’, ‘V’, and ‘T’ represent the acoustic, visual, and textual modalities respectively. ‘U’ indicates a unimodal variant of VAT.**

| A                  | V  | T  | Model    | MER-MULTI<br>Com ( $\uparrow$ ) |
|--------------------|----|----|----------|---------------------------------|
| Unimodal Results   |    |    |          |                                 |
| HB                 | —  | —  | Baseline | 0.24 $\pm$ 0.02                 |
| HL                 | —  | —  | Baseline | 0.36 $\pm$ 0.01                 |
| HB                 | —  | —  | VAT (U)  | 0.58                            |
| HL                 | —  | —  | VAT (U)  | 0.65                            |
| —                  | RF | —  | Baseline | 0.25 $\pm$ 0.01                 |
| —                  | MR | —  | Baseline | 0.19 $\pm$ 0.01                 |
| —                  | RF | —  | VAT (U)  | 0.47                            |
| —                  | MR | —  | VAT (U)  | 0.49                            |
| —                  | —  | ML | Baseline | -0.06 $\pm$ 0.01                |
| Multimodal Results |    |    |          |                                 |
| HL                 | RF | —  | Baseline | 0.51 $\pm$ 0.01                 |
| HL                 | MR | —  | Baseline | 0.53 $\pm$ 0.01                 |
| HL                 | RF | ML | Baseline | 0.56 $\pm$ 0.00                 |
| HL                 | RF | —  | VAT      | 0.69                            |
| HL                 | MR | —  | VAT      | <b>0.70</b>                     |

weighted average F-score is employed as the evaluation metric, while for the dimensional emotion, the mean squared error (MSE) is used as the evaluation metric. Higher F1-score and lower MSE indicate better performance. To evaluate the comprehensive performance, a combined metric Com on discrete and dimension predictions is defined as:  $\text{Com} = \text{F1-Score} - 0.25 * \text{MSE}$ .

### 3.2 Implementation Details

All experiments were conducted using a computational setup comprising eight NVIDIA V100 GPUs, each equipped with 32GB of memory. The CPUs utilized were Intel® Xeon® Gold 6148 Processors with a cache size of 27.5M and a clock speed of 2.40 GHz. The Adam [16] optimizer with a weight decay of 0.02 was employed for optimization purposes. The learning rate was initialized to  $1e-5$  and gradually warmed up to  $1e-4$  after 10 training epochs. To mitigate overfitting, a dropout rate of 0.25, as proposed by Srivastava *et al.* [26], was applied. Subsequently, the learning rate was reduced according to a cosine decay schedule until it reached  $1e-5$ . Early stopping criteria were implemented, whereby if the F1-score on the validation set did not improve for 10 consecutive epochs, the training process was terminated. The video encoder consisted of 6 layers, while the audio encoder comprised 24 layers. For multimodal fusion, the default configuration employed 4 transformer blocks. Furthermore, the model weights pretrained in the MER-SEMI [20] task were utilized to initialize the entire model.

**Table 3: Performance comparison between baseline models and VAT in unimodal and multimodal settings. Here, ‘A’, ‘V’, and ‘T’ indicate the acoustic, visual, and textual modalities respectively. ‘U’ indicates a unimodal variant of VAT.**

| A                  | V  | T  | Model    | MER-NOISE<br>Com ( $\uparrow$ ) |
|--------------------|----|----|----------|---------------------------------|
| Unimodal Results   |    |    |          |                                 |
| HB                 | —  | —  | Baseline | -0.35 $\pm$ 0.04                |
| HL                 | —  | —  | Baseline | 0.16 $\pm$ 0.01                 |
| HB                 | —  | —  | VAT (U)  | 0.55                            |
| HL                 | —  | —  | VAT (U)  | 0.63                            |
| —                  | RF | —  | Baseline | 0.23 $\pm$ 0.01                 |
| —                  | MR | —  | Baseline | 0.18 $\pm$ 0.00                 |
| —                  | RF | —  | VAT (U)  | 0.41                            |
| —                  | MR | —  | VAT (U)  | 0.45                            |
| —                  | —  | RL | Baseline | -0.08 $\pm$ 0.00                |
| Multimodal Results |    |    |          |                                 |
| HL                 | RF | —  | Baseline | 0.38 $\pm$ 0.00                 |
| HL                 | MR | —  | Baseline | 0.37 $\pm$ 0.01                 |
| HL                 | RF | RL | Baseline | 0.41 $\pm$ 0.01                 |
| HL                 | RF | —  | VAT      | 0.65                            |
| HL                 | MR | —  | VAT      | <b>0.68</b>                     |

### 3.3 Results and Analysis

Table 2 and Table 3 present a comprehensive performance comparison between the baseline model and our proposed VAT in both unimodal and multimodal settings. We carefully select acoustic features from HUBERT-large (HL) [13] and HUBERT-base (HB) [13], as well as visual features from ResNet-FER2013 (RF) [10, 11] and MANet-RAFDB (MR) [18, 35] as the single-modal representations for comparative analysis. It is seen that, in the unimodal setting, our VAT consistently outperforms the baseline model in both visual and acoustic modalities. Specifically, for the acoustic modality, VAT (HL) achieves an impressive F1-score of 0.65, while for the visual modality, VAT (MR) achieves an F1-score of 0.49 in the MER-MULTI challenge. Similar trends of improved performance can be observed in the multimodal scenarios. Notably, in the challenging MER-NOISE task, our VAT (HL, MR) attains the highest F1-scores of 0.68, highlighting the effectiveness and robustness of our model in modeling both visual and acoustic features and capturing their intricate interactions.

## 4 CONCLUSION

In this paper, we present our solutions for recognizing both discrete and dimensional emotions in the sub-challenges of the Multimodal Emotion Recognition Challenge (MER 2023). To mitigate the risk of underfitting individual modalities during multimodal training, we utilize joint training with weighted blending of supervision signals. Moreover, we employ diverse data augmentation techniques at different levels, including the *image*, *waveform*, and *spectrogram* levels, to enhance the robustness of our model. Experimental results fully demonstrate the effectiveness of our approach. Remarkably, our approach achieves the top rank in both the MER-MULTI (0.7005) and MER-NOISE (0.6846) sub-challenges.

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